

1 Integral Equations for Scattering

We now connect the scattering problem to the perturbation theory we developed in previous sections. The time-independent Schrödinger equation for the full problem is

$$(\hat{H}_0 + \hat{V})|\psi\rangle = E|\psi\rangle, \quad \hat{H}_0 = \frac{\hat{p}^2}{2m}, \quad (1)$$

where $\hat{V}$ is the time-independent scattering potential and $E = k^2/(2m)$ is the energy. Rearranging Eq. (1):

$$(E - \hat{H}_0)|\psi\rangle = \hat{V}|\psi\rangle. \quad (2)$$

With this form in mind, we would like to invert the operator $(E - \hat{H}_0)$ to solve for $|\psi\rangle$. Formally, we can write $|\psi\rangle = (E - \hat{H}_0)^{-1} \hat{V}|\psi\rangle$. In addition, we note that for a plane wave $|\phi\rangle$ with energy E , we have $(E - \hat{H}_0)|\phi\rangle = 0$, so we can always add $|\phi\rangle$ to the solution without changing the left-hand side of Eq. (2). This gives us the ansatz

$$|\psi\rangle = |\phi\rangle + (E - \hat{H}_0)^{-1} \hat{V}|\psi\rangle. \quad (3)$$

While formally correct, this expression is not very useful because we do not know how to evaluate the inverse operator $(E - \hat{H}_0)^{-1}$ or $|\psi\rangle$ on the right-hand side either. This looks a lot like some of the starting points in perturbation theory. This is the starting point for the formalism in scattering theory. To put it in some nice mathematical language, we will go back to our discussion of Green's functions (unsurprisingly! Green's functions invert differential operators) and see how they allow us to apply the ideas we developed in perturbation theory to solve for $|\psi\rangle$ iteratively (the Born series).

1.1 Green's Functions

First, recall the propagator is defined as

$$K(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) = \langle \mathbf{x}_1 | \hat{U}(t_1; t_0) | \mathbf{x}_0 \rangle = \langle \mathbf{x}_1 | e^{-i\hat{H}_0(t_1-t_0)} | \mathbf{x}_0 \rangle. \quad (4)$$

Some textbooks call the time evolution operator the propagator, but here we keep them distinct: $\hat{U}(t_1; t_0)$ is the operator and the propagator $K(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0)$ is its position-space matrix element. We defined the retarded and advanced propagators as

$$K_{R,A}(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) = \theta(\pm(t_1 - t_0))K(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0). \quad (5)$$

These helped us invert the time-dependent Schrödinger equation, which in position space reads

$$\left(i \frac{\partial}{\partial t_1} - \hat{H}_0(\mathbf{x}_1) \right) \frac{K_R(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0)}{i} = \delta^{(3)}(\mathbf{x}_1 - \mathbf{x}_0) \delta(t_1 - t_0), \quad (6)$$

$$\left(i \frac{\partial}{\partial t_1} - \hat{H}_0(\mathbf{x}_1) \right) \frac{K_A(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0)}{i} = \delta^{(3)}(\mathbf{x}_1 - \mathbf{x}_0) \delta(t_1 - t_0). \quad (7)$$

From those, we had defined the Green's function $G(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0)$ from the retarded propagator. Now, let us define two kinds of Green's function, $G^{(+)}$ and $G^{(-)}$, as

$$G^\pm(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) = \frac{\theta(\pm(t_1 - t_0))}{i} K(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0), \quad (8)$$

where the $+$ sign corresponds to the retarded (causal, forward in time) propagator and the $-$ sign corresponds to the advanced (backward in time) propagator.

Recall the physical interpretation. Taking the time evolution

$$\langle \mathbf{x}_1 | \psi(t_1) \rangle = \langle \mathbf{x}_1 | \hat{U}(t_1; t_0) | \psi(t_0) \rangle, \quad (9)$$

and inserting a complete set of position states, we have

$$\psi(\mathbf{x}_1, t_1) = \int d^3x_0 K(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) \psi(\mathbf{x}_0, t_0). \quad (10)$$

The propagator K gives the amplitude for a particle to propagate from $\mathbf{x}_0$ at time t_0 to $\mathbf{x}_1$ at time t_1 . If we start with $\psi(\mathbf{x}_0, t_0) = \delta^{(3)}(\mathbf{x}_0 - \mathbf{x}')$, then $\psi(\mathbf{x}_1, t_1) = K(\mathbf{x}_1, t_1; \mathbf{x}', t_0)$ is the amplitude of the wavefunction to propagate from $\mathbf{x}'$ at time t_0 to $\mathbf{x}_1$ at time t_1 . This is analogous to what you saw in electrodynamics: “the Green’s function gives the response of the system to a point charge,” except here the “source” is a point-like initial wavefunction.

The propagator K has a spectral representation in terms of the energy eigenstates of our system, $|n\rangle$, with eigenvalues E_n :

$$K(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) = \sum_n \langle \mathbf{x}_1 | n \rangle e^{-iE_n(t_1-t_0)} \langle n | \mathbf{x}_0 \rangle = \sum_n e^{-iE_n(t_1-t_0)} \psi_n(\mathbf{x}_1) \psi_n^*(\mathbf{x}_0), \quad (11)$$



Derive the free-particle propagator

$$K_0(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) = (m/2\pi i(t_1 - t_0))^{3/2} \exp \left[i \frac{m(\mathbf{x}_1 - \mathbf{x}_0)^2}{2(t_1 - t_0)} \right], \quad (12)$$

by using Eq. (11).



Using $\psi_{\mathbf{k}}(\mathbf{x}) = \langle \mathbf{x} | \mathbf{k} \rangle = e^{i\mathbf{k} \cdot \mathbf{x}} / (2\pi)^{3/2}$ and defining $t = t_1 - t_0$ and $\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_0$:

$$K_0(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} e^{-i\frac{k^2}{2m}t} e^{i\mathbf{k} \cdot \mathbf{r}} \quad (13)$$

where the sum over discrete states was replaced by an integral over the continuum of momentum states and we used $E_k = k^2/(2m)$. Completing the square in the exponent, we get a Gaussian integral:

$$\begin{aligned} \int d^3\mathbf{k} e^{-i\frac{k^2}{2m}t} e^{i\mathbf{k} \cdot \mathbf{r}} &= e^{i\frac{m\mathbf{r}^2}{2t}} \int d^3\mathbf{k} e^{-i\frac{t}{2m}(\mathbf{k} - \frac{m\mathbf{r}}{t})^2} \\ &= e^{i\frac{m\mathbf{r}^2}{2t}} \left(\frac{2\pi m}{it} \right)^{3/2} = (m/2\pi it)^{3/2} \exp \left[i \frac{m\mathbf{r}^2}{2t} \right], \end{aligned} \quad (14)$$

where $r^2 = \mathbf{r} \cdot \mathbf{r} = (\mathbf{x}_1 - \mathbf{x}_0)^2$. Now, substituting back into the expression for K_0 :

$$\begin{aligned} K_0(\mathbf{x}_1, t_1; \mathbf{x}_0, t_0) &= \int \frac{d^3\mathbf{k}}{(2\pi)^3} e^{-i\frac{k^2}{2m}(t_1-t_0)} e^{i\mathbf{k} \cdot (\mathbf{x}_1 - \mathbf{x}_0)} \\ &= \left(\frac{m}{2\pi i(t_1 - t_0)} \right)^{3/2} \exp \left[i \frac{m(\mathbf{x}_1 - \mathbf{x}_0)^2}{2(t_1 - t_0)} \right]. \end{aligned} \quad (15)$$

Now we use the spectral decomposition to move from the time domain to the energy domain, which is the natural setting for scattering problems (we are interested in asymptotic states with definite energy, not necessarily in the full time evolution). In particular, at this point, we will assume that the Hamiltonian is time-independent, so that the only dependence on time enters as the difference $t = t_1 - t_0$. We will define the energy-domain Green’s function $G^{(+)}$ as the Fourier transform of the retarded propagator:

$$G^{(+)}(\mathbf{x}_1, \mathbf{x}_0; E) = \frac{1}{i} \int_{-\infty}^{\infty} dt e^{iEt} K_R(\mathbf{x}_1, \mathbf{x}_0; t) = \frac{1}{i} \int_0^{\infty} dt e^{iEt} K(\mathbf{x}_1, \mathbf{x}_0; t), \quad (16)$$

where $t = t_1 - t_0$ and the $\theta(t)$ inside K_R restricts the integral to $t > 0$. Inserting the spectral decomposition Eq. (11):

$$G^{(+)}(\mathbf{x}_1, \mathbf{x}_0; E) = \frac{1}{i} \sum_n \psi_n(\mathbf{x}_1) \psi_n^*(\mathbf{x}_0) \int_0^\infty dt e^{i(E-E_n)t}. \quad (17)$$

This integral does *not* converge as written since the integrand oscillates forever. Still, we insist on this form by manually adding a convergence factor $e^{-\epsilon t}$ with small ϵ that can be sent to 0^+ at the end. This prescription dampens the integrand at large t so that

$$\int_0^\infty dt e^{i(E-E_n+i\epsilon)t} = \left[\frac{e^{i(E-E_n+i\epsilon)t}}{i(E-E_n+i\epsilon)} \right]_0^\infty = \frac{i}{E-E_n+i\epsilon}, \quad (18)$$

where the upper limit vanishes because $e^{-\epsilon t} \rightarrow 0$ as $t \rightarrow \infty$. The $+i\epsilon$ prescription will be soon shown to be related to the boundary condition of scattering problems. Substituting back, we obtain the spectral representation of the energy **Green's function**:

$$G^{(+)}(\mathbf{x}_1, \mathbf{x}_0; E) = \sum_n \frac{\psi_n(\mathbf{x}_1) \psi_n^*(\mathbf{x}_0)}{E - E_n + i\epsilon}, \quad (19)$$

or in terms of the **Green's operator** $\hat{G}^{(+)}(E)$:

$$\hat{G}^{(+)}(E) = \frac{1}{E - \hat{H}_0 + i\epsilon} = \sum_n \frac{|n\rangle \langle n|}{E - E_n + i\epsilon}. \quad (20)$$

The continuum versions are analogous. We could have followed a similar procedure for the advanced Green's function $G^{(-)}(E)$, obtained from $K_A = \theta(-t) K$. In that case, convergence of the integral as $t \rightarrow -\infty$ would require a damping factor $e^{+\epsilon t}$ (with $\epsilon > 0$), which shifts the energy as $E \rightarrow E - i\epsilon$ and yields the $-i\epsilon$ prescription. So, overall, we *define*

$$\hat{G}^{(\pm)}(E) = \frac{1}{E - \hat{H}_0 \pm i\epsilon}, \quad (21)$$

though in non-relativistic QM, only the retarded Green's function $G^{(+)}$ is relevant for scattering.

Now, let us keep ϵ around but forget where it came from. You can ask, given Eq. (21), what is the physical meaning of the $+i\epsilon$ prescription without referring to the time-domain? It turns out you can understand the $+i\epsilon$ prescription as a physical boundary condition in scattering problems. Let us see how this comes about with the free-particle Green's function.

Free-particle Green's function. Let us now evaluate Eq. (19) for a free plane wave using the same notation as before. Using $E = k^2/2m$ and $\psi_{\mathbf{k}}(\mathbf{x}) = e^{i\mathbf{k}\cdot\mathbf{x}}/(2\pi)^{3/2}$, we have

$$G_0^{(+)}(\mathbf{x}_1, \mathbf{x}_0; E) = \int \frac{d^3\mathbf{k}'}{(2\pi)^3} \frac{e^{i\mathbf{k}'\cdot\mathbf{r}}}{E - k'^2/(2m) + i\epsilon} = 2m \int \frac{dk' d\Omega_{\mathbf{k}'}}{(2\pi)^3} \frac{k'^2 e^{ik'r \cos \theta}}{k^2 - k'^2 + i\epsilon'}, \quad (22)$$

where we wrote $\mathbf{k}' \cdot \mathbf{r} = k' r \cos \theta$ with $k' = |\mathbf{k}'|$ and defined $\epsilon' = \epsilon/(2m)$. Evaluating the angular integrals is now straightforward,

$$G_0^{(+)} = \frac{2m}{(2\pi)^2} \int_0^\infty dk' \frac{k'^2}{k^2 - k'^2 + i\epsilon'} \frac{2 \sin(k' r)}{k' r}. \quad (23)$$

Now comes the trick: the integrand is even in k' , so we can extend the integral to the entire real line $(-\infty, \infty)$ and divide by 2. Next, we write $\sin(k' r) = (e^{ik'r} - e^{-ik'r})/(2i)$. The two integrals corresponding to the two exponentials are actually the same after a changing variables

$k' \rightarrow -k'$ in the second integral, so we can just keep one of them and multiply by 2. Overall, the final expression is equivalent to taking $\sin(k'r)$ to $e^{ik'r}/i$, so

$$G_0^{(+)} = \frac{-2m}{(2\pi)^2 i r} \int_{-\infty}^{\infty} dk' \frac{k' e^{ik'r}}{k'^2 - k^2 - i\epsilon'} = \frac{-2m}{(2\pi)^2 i r} \int_{-\infty}^{\infty} dk' \frac{k' e^{ik'r}}{(k' - k - i\epsilon')(k' + k + i\epsilon')}, \quad (24)$$

where we factored the denominator in the second line. The poles of the integrand are $k' = \pm(k + i\epsilon')$, which lie just above and just below the real axis, respectively.



Please review the **complex analysis toolkit** if you do not follow the steps below.

We can use Cauchy's residue theorem to evaluate the integral by promoting k' to a complex variable $k' \rightarrow \kappa$ and integrating along the real axis. Because the factor $e^{i\kappa r}$ decays to zero in the upper half-plane as $|\kappa| \rightarrow \infty$, the real axis can be closed with a semicircle in the upper half-plane without changing the value of the integral (by Jordan's lemma)¹. This semicircular contour on the upper plane encloses the pole of the propagator at $\kappa = +k + i\epsilon'$ and the contour integral is then evaluated using the residue theorem:

$$\int_{-\infty}^{\infty} d\kappa \frac{\kappa e^{i\kappa r}}{(\kappa - k - i\epsilon')(\kappa + k + i\epsilon')} = 2\pi i \text{Res} \left[\frac{\kappa e^{i\kappa r}}{(\kappa - k - i\epsilon')(\kappa + k + i\epsilon')}, \kappa = k + i\epsilon' \right] \quad (25)$$

$$= 2\pi i \lim_{\kappa \rightarrow k + i\epsilon'} (\kappa - (k + i\epsilon')) \frac{\kappa e^{i\kappa r}}{(\kappa - k - i\epsilon')(\kappa + k + i\epsilon')} \quad (26)$$

$$= \pi i e^{i(k + i\epsilon')r}. \quad (27)$$

The integral therefore simplified dramatically and the final result for our free-particle energy domain Green's function is

$$G_0^{(+)}(\mathbf{x}_1, \mathbf{x}_0; E) = -\frac{2m}{4\pi} \frac{e^{ikr}}{r}, \quad (28)$$

where $r = |\mathbf{x}_1 - \mathbf{x}_0|$ and $k = \sqrt{2mE}$. This result is quite impressive.



Physical interpretation: The $+i\epsilon$ prescription selected the pole in the upper half of the complex κ -plane, yielding the outgoing spherical wave e^{+ikr}/r . Had we used $-i\epsilon$ (the advanced Green's function), the contour would enclose the pole at $\kappa = -k - i\epsilon'$ instead, giving the incoming wave e^{-ikr}/r . This is remarkable: the retarded Green's function $G^{(+)}$ automatically produces outgoing spherical waves, while the advanced Green's function $G^{(-)}$ produces incoming ones.

Where are these spherical waves coming from? Recall from Eq. (10) that if we start with a pinched initial wavefunction $\psi(\mathbf{x}_0, t_0) = \delta^{(3)}(\mathbf{x}_0 - \mathbf{x}')$, the time-domain propagator gives $\psi(\mathbf{x}_1, t_1) = K(\mathbf{x}_1, t_1; \mathbf{x}', t_0)$: the point-like disturbance spreads out as a wave. The energy-domain Green's function $G_0^{(+)}(\mathbf{x}_1, \mathbf{x}'; E)$ is the Fourier transform of this causal evolution for a state of definite energy E . The causal response to a point source at $\mathbf{x}'$ is an outgoing spherical wave e^{ikr}/r centered on $\mathbf{x}'$. Using the advanced Green's function instead would give an incoming wave e^{-ikr}/r converging onto $\mathbf{x}'$, which clearly is unphysical for a scattering problem.

¹Recall that $\int_{-\infty}^{\infty} f(x)dx = \int_{\mathcal{C}} f(z)dz - \int_{C_R} f(z)dz$, where $\mathcal{C}$ is the closed contour along the real axis and C_R is the semicircle in the upper half-plane. If $f(z)$ decays sufficiently fast at infinity, then $\int_{C_R} f(z)dz \rightarrow 0$ as $R \rightarrow \infty$, so $\int_{-\infty}^{\infty} f(x)dx = \int_{\mathcal{C}} f(z)dz$. Finally, by the residue theorem, the contour integral $\int_{\mathcal{C}} f(z)dz$ is equal to $2\pi i$ times the sum of the residues of $f(z)$ at the poles enclosed by $\mathcal{C}$.

1.2 The Lippman-Schwinger Equation

Let us now return to Eq. (2). The language of Green's functions allows us to invert differential operators like $(E - \hat{H}_0)$, which is what we need to solve for $|\psi\rangle$. Indeed, the Green's operator $\hat{G}_0^{(+)}(E) = (E - \hat{H}_0 + i\epsilon)^{-1}$ is precisely the inverse of $(E - \hat{H}_0)$ with desired property of producing outgoing waves, which we already argued is the correct boundary condition for scattering problems. Therefore, we can write the solution to Eq. (2) as

$$|\psi^{(+)}\rangle = |\phi\rangle + \hat{G}_0^{(+)}(E) \hat{V} |\psi^{(+)}\rangle, \quad \hat{G}_0^{(+)}(E) = \frac{1}{E - \hat{H}_0 + i\epsilon}. \quad (29)$$

This is the **Lippmann-Schwinger equation**. Note that the state $\psi^{(+)}$ is only determined up to the addition of a state solution of $(E - \hat{H}_0)|\phi\rangle = 0$, which is just a free plane wave with the same energy E (we are always allowed to add zero to the solution). It also carries the superscript $(+)$ to indicate that it satisfies the outgoing wave boundary condition (hence the use of $G_0^{(+)}$ with $+i\epsilon$ prescription).

Next, let us rewrite the Lippmann-Schwinger equation in position space by dotting it with $\langle \mathbf{x} |$, inserting a complete set of position states, and using Eq. (19):

$$\boxed{\psi^{(+)}(\mathbf{x}) = \phi(\mathbf{x}) + \int d^3x' G_0^{(+)}(\mathbf{x}, \mathbf{x}') V(\mathbf{x}') \psi^{(+)}(\mathbf{x}'),} \quad (30)$$

where $\psi^{(+)}(\mathbf{x}) = \langle \mathbf{x} | \psi^{(+)} \rangle$ and $\phi(\mathbf{x}) = \langle \mathbf{x} | \phi \rangle = e^{i\mathbf{k}\cdot\mathbf{x}} / (2\pi)^{3/2}$. As such, the Lippmann-Schwinger equation is an integral equation for the scattering state $|\psi^{(+)}\rangle$, which can be solved iteratively as in the Born series that we see below.



We should have seen this coming from the Green's function language: at each point $\mathbf{x}'$, the potential $V(\mathbf{x}')$ acts as a source, radiating an outgoing spherical wave $G_0^{(+)}(\mathbf{x}, \mathbf{x}') \propto e^{ik|\mathbf{x}-\mathbf{x}'|}/|\mathbf{x}-\mathbf{x}'|$ with amplitude proportional to $V(\mathbf{x}') \psi^{(+)}(\mathbf{x}')$. The full solution is then the superposition of the incoming plane wave $\phi(\mathbf{x})$ and all these outgoing spherical waves from each point in space where the potential is nonzero.

This is analogous to Huygens' principle in optics. Every point on a wavefront acts as a source of secondary spherical waves, and the new wavefront is the envelope of all these spherical waves. The potential here plays the role of the scattering sites in the optics analogy.

Recall the ansatz we took for the scattering state $|\psi\rangle$ in our previous notes:

$$\psi^{(+)}(\mathbf{x}) \sim \frac{1}{(2\pi)^{3/2}} \left[e^{i\mathbf{k}\cdot\mathbf{x}} + f(\theta, \phi) \frac{e^{ikr}}{r} \right], \quad r = |\mathbf{x}|. \quad (31)$$

The overall factor of $(2\pi)^{-3/2}$ is the Dirac normalization of the incident plane wave $\phi(\mathbf{x}) = \langle \mathbf{x} | \mathbf{k} \rangle$ and is carried by the full scattering state; the scattering amplitude $f(\theta, \phi)$ is defined as the coefficient multiplying e^{ikr}/r relative to the incident wave, so the $(2\pi)^{-3/2}$ factor out front is common to both terms and drops out of f .

To extract the scattering amplitude $f(\theta, \phi)$ from the Lippmann-Schwinger equation, we need to analyze the asymptotic behavior of $\psi^{(+)}(\mathbf{x})$ as $r \rightarrow \infty$. Following our list of assumptions, the potential has a finite range, so for sufficiently large r , the potential term in Eq. (30) only receives contributions from $\mathbf{x}'$ within a finite region around the origin. Then, redefining $r = |\mathbf{x}| \gg |\mathbf{x}'|$ (far from the scattering region), we can write

$$|\mathbf{x} - \mathbf{x}'| \simeq r - \hat{\mathbf{r}} \cdot \mathbf{x}', \quad (32)$$

where $\hat{r} = \mathbf{x}/r$ is the unit vector in the observation direction. At such large distances, we are justified in using the *free-particle* Green's function in Eq. (28):

$$G_0^{(+)}(\mathbf{x}, \mathbf{x}') \rightarrow -\frac{2m}{4\pi} \frac{e^{ikr}}{r} e^{-ik\hat{r}\cdot\mathbf{x}'} \quad (33)$$

Substituting back into Eq. (30):

$$\psi^{(+)}(\mathbf{x}) \xrightarrow{r \rightarrow \infty} \frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{(2\pi)^{3/2}} + \left[-\frac{2m}{4\pi} \int d^3x' e^{-i\vec{k}'\cdot\mathbf{x}'} V(\mathbf{x}') \psi^{(+)}(\mathbf{x}') \right] \frac{e^{ikr}}{r}, \quad (34)$$

where $\vec{k}' = k\hat{r}$. Comparing with the asymptotic form in Eq. (31), we identify the incident wave $\phi(\mathbf{x}) = e^{i\mathbf{k}\cdot\mathbf{x}}/(2\pi)^{3/2}$ and the scattering amplitude:

$$f(\theta, \phi) = -\frac{2m(2\pi)^{3/2}}{4\pi} \int d^3x' e^{-i\vec{k}'\cdot\mathbf{x}'} V(\mathbf{x}') \psi^{(+)}(\mathbf{x}'), \quad (35)$$

where $\vec{k}' = k\hat{r}$ is the wavevector of the scattered particle in the direction of observation (with $|\vec{k}'| = k$, since the scattering is elastic).²

Note that this expression is not directly useful because $\psi^{(+)}$ on the right-hand side is not known, $\psi^{(+)}$ is the full scattering solution. While this expression is only valid at large distances and away from the exact forward direction, it makes no approximations regarding the strength of the potential, and that is precisely the clue we need to make progress. Let us now turn to perturbation theory.

1.3 The Born approximation

If the potential $\hat{V}$ is weak, we can solve the Lippmann-Schwinger equation iteratively by expanding $|\psi^{(+)}\rangle$ in powers of $\hat{V}$. This series expansion is called the **Born series** and looks like

$$|\psi^{(+)}\rangle = |\psi^{(0)}\rangle + |\psi^{(1)}\rangle + |\psi^{(2)}\rangle + \dots \quad (36)$$

$$= |\phi\rangle + \hat{G}_0^{(+)} \hat{V} |\phi\rangle + \hat{G}_0^{(+)} \hat{V} \hat{G}_0^{(+)} \hat{V} |\phi\rangle + \dots \quad (37)$$

Clearly, in the absence of a potential, $|\psi^{(+)}\rangle$ reduces to the free plane wave $|\phi\rangle$, as expected. The exact form of the higher-order terms will be shown below, but you should be at least convinced that for a weak potential, the solution cannot be too different from the free plane wave. How such a plane wave gets distorted is exactly what the following procedure captures.

Start with the zeroth-order guess $|\psi^{(+)}\rangle \approx |\phi\rangle$. Substituting $\psi^{(+)}(\mathbf{x}') \approx \phi(\mathbf{x}') = e^{i\mathbf{k}\cdot\mathbf{x}'}/(2\pi)^{3/2}$ into Eq. (35), the factor of $(2\pi)^{3/2}$ in Eq. (35) cancels the $(2\pi)^{-3/2}$ from ϕ , giving

$$f^{(1)}(\theta, \phi) = -\frac{2m}{4\pi} \int d^3x' e^{-i\vec{q}\cdot\mathbf{x}'} V(\mathbf{x}') = -\frac{2m}{4\pi} \tilde{V}(\vec{q}). \quad (38)$$



This approximate equality is called the Born approximation. The dependence on the momentum transfer $\vec{q} = \vec{k}' - \vec{k} = \vec{p}_f - \vec{p}_i$ shows that the scattering amplitude in the Born approximation is proportional to the Fourier transform of the potential, $\tilde{V}(\vec{q}) = \int d^3x e^{-i\vec{q}\cdot\mathbf{x}} V(\mathbf{x})$.

²Rather embarrassingly, the factor of $(2\pi)^{3/2}$ here is a bit non-standard because the scattered wave in Eq. (34) should match the ansatz Eq. (31), which defines f relative to the unit-amplitude incident wave $e^{i\mathbf{k}\cdot\mathbf{x}}$. This factor cancels later on anyway, so if it bothers you, just hang in there.

Higher orders of the Born series are obtained by iterating the Lippmann-Schwinger equation. To get the second-order term, we substitute the first-order solution $|\psi^{(+)}\rangle \approx |\phi\rangle + \hat{G}_0^{(+)} \hat{V} |\phi\rangle$ back into the right-hand side of the Lippmann-Schwinger Eq. (29) to get

$$|\psi^{(2)}\rangle \simeq |\phi\rangle + \hat{G}_0^{(+)} \hat{V} |\psi^{(1)}\rangle = |\phi\rangle + \hat{G}_0^{(+)} \hat{V} (|\phi\rangle + \hat{G}_0^{(+)} \hat{V} |\phi\rangle), \quad (39)$$

such that the second-order scattering amplitude is

$$f^{(2)} = -\frac{2m}{4\pi} \left[\tilde{V}(\vec{q}) + \int \frac{d^3p}{(2\pi)^3} \frac{\tilde{V}(\vec{k}' - \mathbf{p})}{E - p^2/(2m) + i\epsilon} \tilde{V}(\mathbf{p} - \vec{k}) + \dots \right]. \quad (40)$$

Each $\hat{V}$ contributes a Fourier-transformed potential $\tilde{V}$ (a momentum kick), each $\hat{G}_0^{(+)}$ contributes a propagator $1/(E - p^2/(2m) + i\epsilon)$, and the intermediate momentum $\mathbf{p}$ is integrated over. Compare this with the perturbative expansion of the transition amplitude $\mathcal{M}_{fi}$ from our discussion of time-dependent perturbation theory. The Born series is the time-independent counterpart of the Dyson series!



At this point, we could already calculate scattering amplitudes and cross sections for specific potentials like the Yukawa potential, the Coulomb potential, etc. More notes to come on this, but you can already try out $V(r) = -\alpha/r$ and see how the Born rule gives the Rutherford scattering amplitude.

1.4 Momentum-space representation

So far we have written the Lippmann-Schwinger equation in position space. But we already argued that the Green's function is an operator, so we are free to take its matrix elements in any basis. The momentum basis is particularly powerful because momentum eigenstates $|\mathbf{k}\rangle$ are also eigenstates of $\hat{H}_0$, so the free Green's function becomes diagonal.

Using the spectral representation Eq. (19) and inserting momentum eigenstates $|\mathbf{k}\rangle$ with $\hat{H}_0 |\mathbf{k}\rangle = \frac{k^2}{2m} |\mathbf{k}\rangle$:

$$\langle \mathbf{k}_1 | \hat{G}_0^{(+)}(E) | \mathbf{k}_0 \rangle = \frac{\delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_0)}{E - k_0^2/(2m) + i\epsilon}. \quad (41)$$

The delta function enforces $\mathbf{k}_1 = \mathbf{k}_0$: a free particle cannot change its momentum, so the free Green's function is diagonal in momentum space.

The potential matrix element between Dirac-normalized plane waves is

$$\langle \mathbf{k}' | \hat{V} | \mathbf{k} \rangle = \int d^3x \frac{e^{-i\mathbf{k}' \cdot \mathbf{x}}}{(2\pi)^{3/2}} V(\mathbf{x}) \frac{e^{i\mathbf{k} \cdot \mathbf{x}}}{(2\pi)^{3/2}} = \frac{1}{(2\pi)^3} \int d^3x e^{-i\vec{q} \cdot \mathbf{x}} V(\mathbf{x}) = \frac{\tilde{V}(\vec{q})}{(2\pi)^3}, \quad (42)$$

where $\vec{q} = \mathbf{k}' - \mathbf{k}$ is the momentum transfer and $\tilde{V}(\vec{q}) = \int d^3x e^{-i\vec{q} \cdot \mathbf{x}} V(\mathbf{x})$ is the Fourier transform of the potential.

(to be continued... T matrix)